// CROSS AI SOFTWARES INC. · EXECUTIVE ARCHITECTURE

NextGen Architecture **Blueprint**

Residential Property Intelligence Operating System — Stratex Core · Stratex Passport · Stratex Habitat. Version 1.1 executive review package incorporating 20 mandatory corrections.

STATUS · REVISED V1.1 · AWAITING EXECUTIVE APPROVAL

VERSION 1.1 · REVISED February 26, 2026 · CONFIDENTIAL — PROPRIETARY

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STRATEX CORE — NEXTGEN ARCHITECTURE BLUEPRINT v1.1

Executive Summary (One Page)

Classification: CONFIDENTIAL — PROPRIETARY

Status: REVISED · Awaiting Executive Approval

Author: TC · under directive of Anthony Cross

Delivered: Feb 26, 2026

What v1.1 is

A **documentation-only revision pass** applying **all 20 mandatory executive corrections** to Blueprint v1.0. v1.0 is preserved unchanged as read-only reference. No implementation code was written. Production (`stratexdrone.com`) remains untouched.

The four architectural shifts that matter

1. **Model-vendor neutrality** — Architecture references capability interfaces (`ThermalAnalysisProvider`, `MeasurementProvider`, etc.), never vendor names. Language models are confined to narrative composition; deterministic and specialized systems remain authoritative for measurement, safety, and structural decisions.
2. **Scientifically credible thermal registration** — A 10-step calibrated pipeline (extrinsics estimation, reprojection, occlusion testing, per-region residuals) replaces the single-homography approach. Colored thermal texture is never presented as a validated thermal finding.
3. **Secure Passport access** — The permanent hash is not a bearer credential. Access requires expiring signed links + scoped projections + revocation + access logging + rate limiting. Separate projections per audience (public / homeowner / contractor / adjuster / insurer).
4. **Risk-tiered Human QA** — Four tiers replace blanket approval, preventing a review bottleneck while preserving expert sign-off on high-consequence findings.

New sections introduced in v1.1

- **§10** Risk-tiered Human QA (four tiers)
- **§11.2** Property identity resolution (address ≠ identity; identity ≠ ownership)
- **§13** Retention, deletion, legal-hold policy per data class
- **§14** Original sensor data preservation (mission package is source of truth)
- **§16** Product-cost architecture (DayScan · AWE Scan · Elite as first-class products)
- **§17** Formal validation program (15 dimensions) gates marketing language

Deliverables included in this package

1. Full Markdown blueprint v1.1
2. Professional PDF v1.1 (cover · TOC · page numbers · revision history)
3. Redline change log (v1.0 → v1.1)
4. Architecture Decision Records (10 ADRs)
5. This one-page executive summary
6. Preserved v1.0 (read-only)

Approval status

Blueprint v1.0 — approved directionally

Blueprint v1.1 — awaiting explicit executive approval

Phase 1a implementation — **withheld** until v1.1 is signed on Section 23

The single sentence answer to "what changed"

> v1.1 replaces vendor lock-in, over-confident thermal claims, insecure public access, and unscalable human-QA gates with capability interfaces, calibrated thermal registration, signed access links, and risk-tiered review — while preserving every strategic declaration of v1.0.

Files preserved (v1.0 unchanged, chmod 444):

- `/app/memory/_v1.0_PRESERVED_BLUEPRINT.md`
- `/app/memory/_v1.0_PRESERVED_APPENDIX.md`
- `/app/memory/_v1.0_PRESERVED_Blueprint.pdf`

Files added / regenerated (v1.1):

- `/app/memory/NEXTGEN_ARCHITECTURE_BLUEPRINT_v1.1.md`
- `/app/memory/NEXTGEN_ARCHITECTURE_REDLINE_v1.0_to_v1.1.md`
- `/app/memory/NEXTGEN_ARCHITECTURE_ADRs_v1.1.md`
- `/app/memory/NEXTGEN_EXECUTIVE_SUMMARY_v1.1.md` (this file)
- `/app/memory/_blueprint_out/STRATEX_NextGen_Architecture_Blueprint_v1.1.pdf`

Production: Untouched. **Implementation code**: None written.

STRATEX CORE — NEXTGEN ARCHITECTURE BLUEPRINT v1.1

Status: REVISED · Awaiting Executive Approval

Classification: CONFIDENTIAL — PROPRIETARY *(revised per §15 correction)*

Created: Feb 26, 2026 · **Revised**: Feb 26, 2026 · **Approved**: —

Author: TC · Stratex AI Product Manager, under directive of Anthony Cross

Reviewer: Anthony Cross (Executive Architect)

Revision identifier: v1.1 · Directive-driven correction pass

Preserves: v1.0 (frozen, read-only in `/app/memory/_v1.0_PRESERVED_*`)

> **Version 1.1 principle**: Model-vendor names removed from architecture. Thermal registration made scientifically credible. Human-QA tiered. Passport access secured. Performance targets replaced with validation targets. All 20 executive corrections incorporated verbatim.

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0. EXECUTIVE FRAMING

Stratex Core NextGen is the **Residential Property Intelligence Operating System** — an enterprise mission-control platform fusing hardware, specialized computation, and human review across three pillars:

PILLAR	PURPOSE
Stratex Core	Contractor operating system — mission, capture, intelligence, delivery
Stratex Passport	Permanent, hash-chained property intelligence record — the source of truth
Stratex Habitat	Homeowner-facing intelligence platform

The drone is one sensor among many. Specialized computer-vision, photogrammetry, radiometric, and rule-based engines perform the authoritative technical work. General-purpose language models are used **only for narrative composition and orchestration hints**, never as the principal engine for measurement, safety, structural, or diagnostic decisions.

Mandatory workflow chain — no module bypasses this:

```
Mission Control → Mission Planning → Mission Validation → Flight →
Mission Assurance → Capture Validation → Digital Twin Generation →
AI Workforce Processing → Property Intelligence → AWE™ Intelligence →
Human QA → Report Generation → Property Passport Update →
Habitat Synchronization → Customer Delivery
```

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1. APPLICATION STATE MODEL *(unchanged from v1.0)*

STATE	LOCATION	DEPLOY TARGET	MODIFIABLE?
PRODUCTION	`stratexdrone.com`	Frozen	Read-only
NEXTGEN	Preview	Active dev	Yes
LEGACY	`/legacy/*`	Not deployable	Read-only after freeze

Migration is irreversible only after 10 parity gates pass (unchanged from v1.0).

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2. MODULE HIERARCHY *(unchanged)*

14 first-class modules across four layers — Operations · Intelligence · Delivery · Platform. Full list identical to v1.0 §2.

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3. NAVIGATION HIERARCHY

Correction §4 applied — AWE™ nav labels revised to reflect scientifically supportable scope:

```
AWE™ INTELLIGENCE
· Air        (ventilation indicators, intake/exhaust observations,
              exterior air-leakage indicators, condensation-risk
              indicators — NOT indoor air quality)
· Water      (moisture-retention indicators, roof water risk,
              wall retention indicators, thermal moisture proxies)
· Energy     (envelope thermal performance, heat-loss indicators,
              exterior thermal signatures)
· AWE Composite Index™  (calibrated per §20 validation program only)
```

All other nav sections identical to v1.0.

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4. DATABASE DOMAINS

Ten **logical modules** inside a single MongoDB database initially. Only the **Passport domain** extracts to a separate database in Phase 2 (unchanged from v1.0 §4 + Appendix Q10).

New retention + PII classification rules per **§13 (new)** now apply per domain.

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5. AI WORKFORCE ARCHITECTURE — REVISED

5.1 Vendor-neutral provider interfaces *(correction §1)*

Every logical agent binds to **capability interfaces**, not vendor names. Concrete provider mapping lives in deployment config, never in the architecture.

INTERFACE	PURPOSE	CONCRETE IMPLEMENTATIONS MAY INCLUDE
`ReasoningProvider`	Orchestration hints, workflow decisions	LLMs, rule engines
`NarrativeProvider`	Human-readable text composition	LLMs
`VisionProvider`	RGB scene classification, semantic segmentation	CV models, foundation vision models
`SegmentationProvider`	Pixel-level object/roof-plane segmentation	Specialized CV
`ThermalAnalysisProvider`	Radiometric anomaly detection	Specialized thermal CV + calibrated statistics
`MeasurementProvider`	Precise measurement extraction from twin	Deterministic geometry engines
`PhotogrammetryProvider`	Mesh + point-cloud generation	Commercial/open-source photogrammetry stacks
`BimProvider`	CAD/BIM derivative generation	Commercial BIM stacks
`RuleEngine`	Deterministic policies (ATC gates, product routing, tier logic)	In-house rules
`CalibrationProvider`	Sensor calibration + registration math	Deterministic algorithms

Safety-critical, measurement-critical, and engineering-relevant operations are performed by deterministic and specialized systems. LLMs never author these values; they may only compose the narrative that accompanies them.

5.2 Agent-run audit record *(strengthened per §1)*

Every agent run persists to `agent_runs` with **all of the following mandatory fields**:

```
agent_run_id, agent_id, agent_version,
provider_id, provider_model, model_version, prompt_version,
algorithm_id, algorithm_version, runtime_config_hash,
input_hash, output_hash,
started_at, finished_at, cost_usd,
confidence_pct, evidence_ids[], provenance{},
implementation_state, verification_status
```

5.3 Twenty-four logical agents

Same 24-agent logical count as v1.0, mapped to five deployable services (Appendix Q1 unchanged). **Every agent binds to interfaces above** — no vendor names in the registry.

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6. API BOUNDARIES

Five **logical route-prefix planes with distinct middleware chains** (Appendix Q11 unchanged). **Correction §6 applied to Public plane** — see §9 below.

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7. EVENT FLOW & STATE MANAGEMENT

7.1 Frontend state

React Query (server state) + Zustand (small UI state). Adoption cost + rationale in Appendix Q13.

7.2 Critical event delivery — durable *(correction §11)*

Critical events must survive process restart, deploy, network interruption, worker failure, and duplicate delivery. **Phase 1 pattern**: transactional outbox in MongoDB + durable jobs collection with locking and idempotent retries. No external broker required.

Critical event set (durable):

- `finding.approved` → Passport append
- `passport.appended` → Habitat sync
- `habitat.ack_pending`
- `report.generation_pending`
- `evidence.package_finalization_pending`

Non-critical events (FastAPI in-process background tasks are acceptable):

- UI notifications
- Cache warming
- Log aggregation ticks

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8. SYNCHRONIZATION FLOW

Core → Passport → Habitat. Cryptographic receipts (existing cosign SHA-256 pattern) on every accepted append. **Passport concurrency revised per §8 correction** — see §11 below.

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9. SECURITY MODEL — REVISED

9.1 Public passport access *(correction §6)*

The permanent Passport identifier hash is not a bearer credential. Public access requires:

- **Expiring signed links** (default 24h TTL, configurable per recipient class)
- **Narrowly scoped projection** per audience: public / homeowner / contractor / adjuster / insurer
- **Recipient binding** optional (link tied to email or org)
- **Revocation** endpoint + kill-list check on every access
- **Access logging** to `audit_events` on every retrieval
- **Rate limiting** per recipient + per IP
- **No exposure** of private owner name, contact info, claim details, contractor internals, or security data on the public projection

9.2 MFA scope *(correction §7)*

MFA (or a phishing-resistant equivalent — WebAuthn/passkeys preferred) required for:

- CEO · Administrators · Organization owners
- Anyone managing credentials
- Anyone approving Tier 3 or Tier 4 findings (§10 below)
- Adjusters accessing claim evidence
- Engineers/reviewers signing controlled findings
- Users exporting sensitive property packages
- Users changing Passport authority or property ownership
- Users initiating privileged service actions

Step-up authentication required for high-risk actions even inside an active session (approvals, cosigns, ownership transfers, service-key rotation, evidence deletion).

9.3 Roles & authorization

Phase 1: `admin`, `contractor`, `pilot`, `inspector`, `homeowner`, `adjuster`, `engineer_reviewer`, `service`. RBAC + property-level ABAC via `property_org_bindings`. Full detail in Appendix Q9.

9.4 Tenant isolation for object storage *(correction §12)*

The **security outcome required is tenant isolation** — not a specific storage layout. Permitted implementations:

- Separate buckets per tenant
- Shared bucket with tenant-prefixed keys enforced by IAM policy
- Per-object access control
- Separate encryption contexts per tenant
- Signed URLs with tenant-scoped claims
- Service-mediated access

Provider choice, cost, and physical layout are deployment decisions — the architecture specifies the security guarantee, not the implementation.

9.5 Provenance chain

Every value resolves to `{provenance_type, model_confidence, measurement_uncertainty, finding_confidence, verification_status, reviewer, reviewed_at, truth_score_band, implementation_state}`.

9.6 Observability

Sentry, structured logs, OpenTelemetry tracing — **AWAITING INTEGRATION** (Phase 2+).

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10. HUMAN QA — RISK-TIERED *(correction §5)*

Replaces the prior blanket "human gate on every finding" with a four-tier framework:

Tier 1 — Automated informational

Image inventory · coverage status · visible-component counts · low-consequence classifications.

May publish automatically when: input quality passes · confidence ≥ approved threshold · no contradictory evidence · finding clearly labeled as automated.

Tier 2 — Contractor review

Roof damage candidates · material quantities · window/door classifications · maintenance recommendations · estimate components.

Requires: qualified contractor or inspector review before customer delivery.

Tier 3 — High-consequence

Probable moisture intrusion · significant energy-loss claims · structural concerns · insurance-related conclusions · safety issues · repair-release decisions · engineering deviations.

Requires: MFA-authenticated qualified human approval with signed cosign receipt.

Tier 4 — Engineering-controlled

Structural adequacy · code compliance · engineering deviations · repair designs · safety certification · building release decisions.

System must not independently approve. Requires appropriately licensed or authorized professional signature.

Finding schema (mandatory fields)

```
finding_id, risk_tier, required_reviewer_role, approval_status,
verification_method, release_restriction, ...
```

Every publisher (Passport append, report render, Habitat sync) checks `risk_tier` + `approval_status` before releasing.

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11. PROPERTY PASSPORT AUTHORITY — REVISED

Ownership matrix unchanged from v1.0 §2 Appendix Q2 (Passport service alone writes; Core submits deltas; Habitat receives projections).

11.1 Concurrency: optimistic with explicit rebase *(correction §8)*

Replaces the prior "first-writer-wins" language. Workflow:

1. Client re-reads current Passport `seq`
2. Compare submitted delta against accepted intervening changes
3. **Automatically rebase** when non-conflicting
4. **Route materially conflicting findings to Human QA queue** — never silently discard
5. Preserve **both** attempted submissions in the audit trail (`passport_sync_log`)
6. Never overwrite or silently discard accepted history

7. Issue a new signed receipt only after successful append

Losing submissions are queued and surfaced; the losing user is notified with a rebase link.

11.2 Property identity resolution *(correction §14 — new)*

Passport identity is **never** a typed street address alone. Identity = controlled combination of:

- Normalized address (postal + geocoded)
- Parcel identifier (county/jurisdiction)
- Geographic coordinates (lat/lon with precision floor)
- Jurisdiction (city/county/state)
- Unit or structure identifier (for multi-unit buildings)
- Property boundary (polygon)
- Existing Passport records (fuzzy match + human review on conflict)

Identity rules:

- Duplicate Passports for the same home → rejected at mint with a match candidate list surfaced to human review
- Two homes incorrectly merged → separable via SUPERSEDE ledger entries
- Unit-level records not confused with parcel-level records (unit is a distinct identity slot)
- Ownership change is **not** a new property — it is a new record in `owners`, linked to the same Passport
- Address changes (renaming, renumbering) do not break history — Passport identity persists

Property identity ≠ ownership identity. They are distinct concepts modeled in separate collections (`properties`, `owners`, `property_org_bindings`).

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12. DIGITAL TWIN ARCHITECTURE — REVISED

12.1 Layered canonical package *(correction §9)*

The Digital Twin is not one file. It is a **layered evidence package**:

LAYER	CONTENTS
Evidence geometry	Original imagery, telemetry, point clouds, camera poses, control points, calibration data, reconstruction report, residual errors
Measurement geometry	Validated surfaces, coordinate reference, scale constraints, measurement entities, tolerances, confidence, provenance
Presentation geometry	glTF/GLB, Draco compression, KTX2 textures, mobile LODs, thermal + findings overlays
Interchange derivatives	IFC, DXF, PLY / LAS / LAZ / E57 where appropriate, GeoTIFF, report images

Compression, decimation, or presentation optimization must never alter the authoritative measurement record. Presentation geometry is derived from — and always subordinate to — measurement geometry.

IFC and DXF exports are generated only where source information supports those elements. The system does not invent concealed framing, wall assemblies, or structural elements it did not observe.

12.2 Thermal-to-3D registration — calibrated pipeline *(correction §2)*

Replaces the prior "feature-based homography per capture" language, which was insufficient for complex three-dimensional houses.

Canonical pipeline:

1. **Preserve original radiometric thermal files** (see §14 below)
2. **Preserve** camera intrinsics, distortion coefficients, timestamps, pose, gimbal position, range information, environmental metadata
3. **Estimate thermal-camera extrinsics** relative to RGB and the property coordinate system
4. **Register each thermal observation** through calibrated camera projection
5. **Reproject radiometric data** onto validated 3D surfaces
6. **Perform depth and occlusion testing**
7. **Reject** reflected, grazing-angle, poorly registered, and hidden surfaces

8. **Store registration residuals and per-region confidence**
9. **Permit local homography only** as a limited method for verified planar regions
10. **Require human review** where registration error exceeds the approved threshold

A colored thermal texture is never presented as a scientifically validated thermal finding.

12.3 Vendor-neutral processors *(unchanged from v1.0)*

Photogrammetry, CAD/BIM, radiometric processing are all adapter contracts (`PhotogrammetryProvider`, `BimProvider`, `ThermalAnalysisProvider`). Concrete provider is swappable without changing Core.

12.4 Performance validation targets *(correction §10)*

Prior claim of "iPhone 12 60 fps" replaced with **testable acceptance targets, subject to measured validation**:

METRIC	TARGET
Initial useful view time	≤ 4 s on supported devices
Maximum mobile payload (L2)	≤ 25 MB per property
Interaction frame-rate	30–60 fps target during normal interaction, subject to measured validation
Peak memory	≤ 400 MB on iOS Safari
Crash-free session rate	≥ 99% target once validated
Layer-switch latency	≤ 1.5 s target
Progressive-loading	L2 → L1 → L0 on demand
Fallback behavior	2D exhibit + gallery view for unsupported devices
Tested devices/browsers	Documented per release

Nothing is promised as a guarantee until benchmarked on approved test properties.

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13. DATA RETENTION, DELETION & LEGAL HOLD *(correction §13 — new)*

The Passport ledger is permanent. Not all other data is.

Data classes + retention

CLASS	RETENTION
Passport ledger records	Permanent (immutable, chain-verified)
Raw mission evidence (imagery, radiometric files, telemetry)	7 years hot storage, then cold storage indefinitely
Derived models (twin, CAD, thermal reprojections)	3 years hot, regenerable from evidence
Temporary working files (renders, caches)	90 days rolling
Personal information (owner name, contact)	Retained while owner-property binding is active; corrected on request; scrubbed on legal deletion request
Authentication logs	2 years hot, 7 years cold
AI prompts + outputs	2 years hot, 7 years cold (audit-linked)
Customer exports	30 days recoverable
Deleted / corrected ownership data	Corrected via SUPERSEDE ledger entries; historical record preserved unless legal deletion overrides

Policies

- **Customer-requested deletion**: honored per GDPR/CCPA where applicable, with cryptographic proof-of-deletion. Passport chain remains but PII fields are tombstoned.
- **Legal hold**: overrides retention; freezes deletion until released.
- **Insurance / claim preservation**: preserves evidence for claim-window duration + statutory buffer.
- **PII correction**: applied via SUPERSEDE, never destructive.

- **Account closure**: contractor-side data retained under org retention policy; homeowner-side data subject to deletion request.
- **Storage lifecycle**: hot → cold transitions automated per class.
- **Secure deletion**: crypto-shred keys for tombstoned records.
- **Backup retention**: matches primary; encrypted; access-audited.
- **Evidence immutability**: raw sensor data is never modified; corrections are new records.
- **Legal / contract exceptions**: documented per contract, override defaults.

Nothing is claimed to be kept forever except the Passport ledger.

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14. ORIGINAL SENSOR DATA PRESERVATION *(correction §3)*

Normalized radiometric TIFF is **not** the sole thermal source of truth. The **canonical mission package** preserves:

- Original DJI radiometric files (R-JPG or vendor-native format)
- Original RGB files
- Original video where collected
- Flight logs
- Telemetry streams
- Sensor metadata (intrinsics, distortion, calibration)
- Checksums per file
- Calibration data
- Environmental records (temperature, humidity, wind, solar loading, precipitation history)
- Operator notes

Normalized TIFF, PNG, JPEG, glTF, point-cloud, and report derivatives are **generated derivatives** stored alongside the mission package.

Original evidence is **immutable and retained per §13**. The mission package — not a converted image — is the evidence source of truth.

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15. PROVENANCE, CONFIDENCE & VERIFICATION LANGUAGE *(corrections §17 + §18)*

15.1 Distinct fields (correction §17)

"Verified" is a review state, not a confidence value. The finding schema separates:

FIELD	TYPE	PURPOSE	
`provenance_type`	enum	Source lineage (measured, calculated, estimated, imported, AI-assisted, human-entered)	
`model_confidence`	0–100	Per-inference confidence	
`measurement_uncertainty`	± value with unit	Per-measurement error bar	
`finding_confidence`	0–100	Aggregate confidence for a finding	
`verification_status`	enum	Unreviewed / reviewed / verified / rejected / requires_field_verification	
`reviewer`	user_id \	null	Approver identity
`reviewed_at`	timestamp \	null	Approval time
`truth_score_band`	enum (10-pt band)	Property record trust indicator	
`implementation_state`	enum	Feature-state per §7 taxonomy	

Human approval does not automatically raise measurement accuracy. It raises `verification_status`, not `measurement_uncertainty`.

15.2 Provenance chip taxonomy (correction §18)

Chips are separated by category. UI may combine visually; the data model keeps them separate.

Derivation: Measured · Calculated · Estimated · Imported · AI-assisted · Human-entered

Verification: Unreviewed · Reviewed · Verified · Rejected · Requires field verification

Implementation state: Operational · Partially operational · Mocked · Planned · Awaiting credentials · Awaiting hardware · Awaiting validation · Deprecated · Legacy

Evidence availability: Complete · Partial · Missing · Contradictory

15.3 Truth Score (unchanged from Appendix Q4)

10-point bands only, never decimals. Marketing restricted until §20 validation passes.

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16. PRODUCT-COST & PROCESSING-GATE ARCHITECTURE *(correction §19 — new)*

Stratex sells predictable **contractor products**, not token accounting. Three first-class products:

Product definitions

PRODUCT	REQUIRED MISSION TYPE	SENSORS	PROCESSING ENGINES	HUMAN QA TIER	REPORT TEMPLATE	PASSPORT UPDATE	HABITAT ENTITLEMENT
Stratex Core DayScan™	Daytime RGB	RGB · GPS/RTK	Photogrammetry · Measurement · Roof/Window/Door Intelligence	Tier 2	DayScan template	DayScan chain	DayScan projection
Stratex AWE™ Scan	Nighttime, eligibility-gated	Radiometric thermal · reference RGB · environmental sensors	Thermal Analysis · Water/Air/Energy AWE agents · Registration	Tier 3	AWE template	AWE chain	AWE projection
Stratex Elite™ Property Intelligence	Both missions (DayScan + AWE Scan)	All	All + cross-registration + AWE Index™	Tier 3	Elite template	Combined update	Elite projection

Per-product policy fields (mandatory)

- Required capture conditions
- Human QA tier
- Contractor price
- Internal estimated cost (recorded to Cost Ledger)
- Upgrade rules (DayScan → Elite path)
- Rescan rules (failed capture → free rescan window)
- Failed-mission policy (partial-refund vs credit)

Orchestrator responsibilities

The AI orchestrator **selects processing engines according to the purchased product and property complexity** — not by exposing token accounting to the contractor.

Internal compute + provider costs are still recorded per mission in the `cost_ledger` collection for finance and margin management.

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17. FORMAL VALIDATION PROGRAM *(correction §20 — new)*

No measurement, AWE score, defect classifier, or performance claim is promoted to "validated" merely because the software runs.

Field-validation program (Phase 2+)

Covers:

- Known-dimension test structures (in-house calibration properties)
- Independent manual roof measurements (surveyor-grade)
- Survey control (fixed control points on test properties)

- Repeat-flight reproducibility
- Different pilots
- Different weather
- Different roof materials (asphalt shingle, metal, tile, membrane)
- Different house geometries (simple gable → complex hip/valley)
- Thermal false positives (moisture-mimic exclusion, ventilation-shadow rejection)
- Moisture-meter comparison
- Blower-door comparison where applicable
- Interior/exterior thermal comparison
- Ground-truth repair findings (post-teardown reconciliation)
- Processor-to-processor comparison (swap `PhotogrammetryProvider` implementations)
- Measurement-error distributions (published per class)
- Failure-rate reporting

Acceptance thresholds

Defined **before** Stratex uses any of the following marketing phrases:

- "Unmatched accuracy"
- "Certified"
- "Verified"
- "Engineering-grade"
- "Insurance-grade"

Marketing language follows validation evidence. Until validated, product copy uses:

"Prototype scoring — not calibrated to industry standard yet."

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18. VISUAL DESIGN PRINCIPLES *(unchanged from v1.0)*

Mission-control aesthetic. Dark graphite base. Provenance chip taxonomy per §15.2. No generic SaaS.

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19. EXECUTIVE ARCHITECTURE REVIEW *(unchanged from v1.0 §12)*

Strengths, weaknesses, tech debt, risks, missing integrations, performance, security, scalability, recommended improvements — retained from v1.0 with the addition that every v1.1 correction resolves a specific weakness.

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20. PHASE 1 SCOPE CONTROL *(unchanged, reinforced)*

Phase 1 delivers architecture + operating shell. Does **not** claim production-ready photogrammetry, thermal diagnosis, engineering measurement, or automated AWE conclusions. Every screen carrying intelligence content wears a chip from §15.2.

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21. FORBIDDEN LIST (Phase 1 non-scope) *(unchanged)*

- No DJI SDK integration
- No photogrammetry pipeline
- No real radiometric thermal ingest
- No CAD/BIM generator
- No real materials/labor DB

- No Habitat platform itself (only sync surface)
- No SSO
- No live payments checkout
- No multi-tenant migration (Phase 2)

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22. IMPLEMENTATION SEQUENCE *(unchanged from v1.0)*

Phase 1a Legacy Freeze · Phase 1b NextGen Shell · Phase 1c Workspace Stubs · Phase 1d Screenshot + Executive Review Package.

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23. APPROVAL BLOCK — v1.1

FIELD	VALUE
Blueprint version	1.1
Created date	Feb 26, 2026
Revised date	Feb 26, 2026
Approved date	☐ awaiting signature
Author	TC · Stratex AI Product Manager
Reviewer	Anthony Cross (Executive Architect)
Status	REVISED · AWAITING APPROVAL
Revision identifier	v1.1 · Directive-driven correction pass

☐ Version 1.1 reviewed by Anthony Cross

☐ All 20 corrections confirmed incorporated

☐ Executive Architecture Review reviewed

☐ Phase 1 acceptance criteria confirmed

☐ Authorized to proceed to Phase 1a implementation

Signature: ______________________ Date: ______________

APPENDIX — REVISION HISTORY

VERSION	DATE	AUTHOR	CHANGE SUMMARY
1.0	Feb 26, 2026	TC	Initial draft (623 lines)
1.0-app	Feb 26, 2026	TC	Review Appendix (Q1-Q14 · 950 lines)
1.1	Feb 26, 2026	TC	**20 mandatory corrections applied. See STRATEX_Blueprint_v1.1_REDLINE.md for redline. v1.0 preserved unchanged at `/app/memory/_v1.0_PRESERVED_*`**

End of Blueprint v1.1.

STRATEX BLUEPRINT — v1.0 → v1.1 REDLINE CHANGE LOG

Scope: Every substantive change from Blueprint v1.0 to v1.1. v1.0 is preserved verbatim at `/app/memory/_v1.0_PRESERVED_*`.

Summary of Changes (20 mandatory corrections applied)

#	CORRECTION	V1.0 LOCATION	V1.1 LOCATION	DELTA
1	Remove model-vendor hardcoding	§5 registry ("Claude Sonnet" hardcoded)	§5.1 · Provider interfaces added	Vendor names → `ReasoningProvider`, `VisionProvider`, `ThermalAnalysisProvider`, `SegmentationProvider`, `MeasurementProvider`, `NarrativeProvider`, `RuleEngine`, `PhotogrammetryProvider`, `BimProvider`, `CalibrationProvider`
2	Thermal-to-3D registration	§6 Digital Twin ("feature-based homography per capture")	§12.2 · calibrated pipeline	10-step calibrated registration replaces single homography; residuals + occlusion + human review on threshold breach
3	Preserve original sensor data	§6 (thermal TIFF as source)	§14 · Canonical mission package	Original DJI radiometric + RGB + video + flight logs + telemetry + calibration are the source of truth; derivatives labeled as such
4	Correct Air Intelligence claims	§3 nav ("air quality, leakage")	§3 nav · scoped labels	Removed indoor air quality claims; scope narrowed to exterior/probable indicators; explicit "mold-risk indicator is not a mold diagnosis"
5	Risk-tiered Human QA	§5 (blanket "Yes" gates)	§10 · four-tier framework	Tier 1 (auto) / Tier 2 (contractor) / Tier 3 (MFA-signed) / Tier 4 (licensed professional); finding schema adds `risk_tier`, `required_reviewer_role`, `release_restriction`
6	Public Passport access	§6 (hash-only public read)	§9.1 · signed-link scheme	Expiring signed links · scoped projections · revocation · access log · rate limits; hash is not a bearer credential
7	MFA scope	§9 (CEO + admin only)	§9.2 · expanded matrix	MFA required for Tier 3+ approvals, adjuster access, engineer reviewers, exports, ownership changes; step-up auth for in-session high-risk actions; WebAuthn/passkeys preferred
8	Passport concurrency	§8 ("first-writer-wins")	§11.1 · optimistic + explicit rebase	7-step rebase workflow; losing submissions routed to Human QA queue; never silently discarded; both attempts preserved in audit
9	Digital Twin source-of-truth	§6 (glTF canonical)	§12.1 · layered package	Four layers: Evidence · Measurement · Presentation · Interchange. Compression never alters measurement record. IFC/DXF only where source supports
10	Performance promises	§6 ("iPhone 12 · 60 fps")	§12.4 · validation targets	9-metric target table with "subject to measured validation" language
11	Durable delivery	§7 (in-process background tasks)	§7.2 · transactional outbox	Critical events on durable jobs collection with idempotent retries; non-critical events remain in-process
12	Asset-storage isolation language	§9 (per-org buckets)	§9.4 · outcome-driven	Isolation is the guarantee; implementation is deployment-decided (buckets · prefixes · IAM · encryption contexts · signed URLs)
13	Retention, deletion, legal hold	none in v1.0	§13 · new section	Per-class retention table + GDPR/CCPA deletion + legal hold + PII correction + secure delete
14	Property identity resolution	none in v1.0	§11.2 · new section	Identity = normalized address + parcel + geo + jurisdiction + unit + boundary; property ≠ ownership; duplicate prevention with human review
15	Remove "BLACK-OPS" label	Cover PDF classification	Cover (v1.1) · replaced	Classification: **CONFIDENTIAL — PROPRIETARY**
16	Document date	PDF cover ("July 18, 2026")	Cover (v1.1) · corrected	Created / Revised / Approved dates explicit; approval date remains blank until signed
17	Provenance / confidence / verification	§5 (confidence 0-100 or "verified" mixed)	§15.1 · distinct fields	9 separate fields: `provenance_type`, `model_confidence`, `measurement_uncertainty`, `finding_confidence`, `verification_status`, `reviewer`, `reviewed_at`, `truth_score_band`, `implementation_state`
18	Provenance chip taxonomy	§11 (single flat list)	§15.2 · categorical	Four separate categories: Derivation · Verification · Implementation state · Evidence availability
19	Product-cost architecture	none in v1.0	§16 · new section	Three products (DayScan · AWE Scan · Elite) with per-product processing/QA/cost policies; token accounting hidden from contractor
20	Formal validation program	none in v1.0	§17 · new section	15-dimension field-validation program; acceptance thresholds gate marketing language

Line-by-line changes (major inserts)

New sections added in v1.1

- **§5.1** Vendor-neutral provider interfaces
- **§5.2** Strengthened agent-run audit record fields
- **§7.2** Durable delivery for critical events
- **§9.1** Public passport access (signed links)
- **§9.2** MFA scope expansion
- **§9.4** Tenant-isolation outcome vs implementation
- **§10** Risk-tiered Human QA (four tiers)
- **§11.1** Optimistic concurrency with explicit rebase
- **§11.2** Property identity resolution
- **§12.1** Layered canonical Digital Twin package
- **§12.2** Calibrated thermal-to-3D registration pipeline
- **§12.4** Validation targets (not guarantees)
- **§13** Data retention, deletion, legal hold
- **§14** Original sensor data preservation
- **§15.1** Distinct provenance / confidence / verification fields
- **§15.2** Categorical provenance chip taxonomy
- **§16** Product-cost & processing-gate architecture
- **§17** Formal validation program
- **§23** Approval block reworked with revision metadata

Sections unchanged from v1.0

- §1 Application State Model
- §2 Module Hierarchy
- §4 Database Domains
- §6 API Boundaries (structure)
- §8 Synchronization Flow (structure)
- §18 Visual Design Principles
- §19 Executive Architecture Review
- §20 Phase 1 Scope Control
- §21 Forbidden List
- §22 Implementation Sequence

Language removals

- "Claude Sonnet 4.6" — every instance replaced with capability interface
- "Feature-based homography" — replaced with calibrated pipeline (§12.2)
- "first-writer-wins" — replaced with optimistic + rebase (§11.1)
- "Human gate: Yes" blanket — replaced with `risk_tier` (§10)
- "iPhone 12 · 60 fps" — replaced with validation-target table (§12.4)
- "BLACK-OPS" classification — replaced with CONFIDENTIAL — PROPRIETARY
- "confidence 0-100 or verified" — split into 9 separate fields (§15.1)

Files changed to create v1.1

Documentation-only. No implementation code touched. No production changes.

```
CREATED  /app/memory/NEXTGEN_ARCHITECTURE_BLUEPRINT_v1.1.md   (this pass)
CREATED  /app/memory/NEXTGEN_ARCHITECTURE_REDLINE_v1.0_to_v1.1.md   (this file)
CREATED  /app/memory/NEXTGEN_ARCHITECTURE_ADRs_v1.1.md
CREATED  /app/memory/NEXTGEN_EXECUTIVE_SUMMARY_v1.1.md
UPDATED  /app/memory/_build_blueprint_pdf.py  (parameterized for v1.1 title + classification)
UPDATED  /app/backend/routes/blueprint.py     (added v1.1 endpoints; v1.0 endpoints preserved)
PRESERVED /app/memory/_v1.0_PRESERVED_BLUEPRINT.md    (read-only, chmod 444)
PRESERVED /app/memory/_v1.0_PRESERVED_APPENDIX.md     (read-only, chmod 444)
PRESERVED /app/memory/_v1.0_PRESERVED_Blueprint.pdf   (read-only, chmod 444)
GENERATED /app/memory/_blueprint_out/STRATEX_NextGen_Architecture_Blueprint_v1.1.pdf
```

Confirmations

- ✅ **Version 1.0 preserved unchanged** — three read-only files at `/app/memory/_v1.0_PRESERVED_*`
- ✅ **Production untouched** — `stratexdrone.com` still runs the last approved build
- ✅ **No NextGen implementation code written** — this pass is documentation-only
- ✅ **Every correction 1–20 incorporated** in v1.1 as tabulated above

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Items the agent could NOT confidently implement in v1.1 (transparent statement)

None. All 20 corrections were within scope of documentation revision. The following items are correctly deferred to later phases and are labeled as such in v1.1:

- Actual DJI SDK adapter code (deferred to Phase 5)
- Concrete provider mappings to `PhotogrammetryProvider` / `BimProvider` (deferred to procurement decisions)
- Live field-validation acceptance thresholds (deferred to §17 execution)
- Live MFA implementation details (deferred to Phase 2 security refactor)
- Concrete numerical Truth Score coefficients (deferred to Phase 2 calibration)

These are **explicitly labeled as awaiting validation / awaiting hardware / awaiting integration** in v1.1 — not glossed over.

STRATEX BLUEPRINT — ARCHITECTURE DECISION RECORDS (ADRs)

Companion to Blueprint v1.1 · Each ADR justifies one major correction.

ADR-001 · Model-Vendor Neutrality

Status: Accepted (v1.1)

Context: Blueprint v1.0 hardcoded "Claude Sonnet 4.6" into agent registry, creating vendor lock-in and incorrectly treating a general-purpose LLM as the principal engine for measurement and safety-critical operations.

Decision: Architecture references only capability interfaces (`ReasoningProvider`, `VisionProvider`, `ThermalAnalysisProvider`, `SegmentationProvider`, `MeasurementProvider`, `NarrativeProvider`, `RuleEngine`, `PhotogrammetryProvider`, `BimProvider`, `CalibrationProvider`). Concrete vendor + model bindings live in deployment configuration only.

Consequences: Provider swap is a config change, not a code change. LLM outputs are confined to narrative composition; deterministic and specialized systems remain authoritative for measurements, safety, and structural findings. Every `agent_runs` row records the actual provider + model + prompt version at runtime for audit.

ADR-002 · Calibrated Thermal Registration Pipeline

Status: Accepted (v1.1)

Context: Feature-based homography per capture is insufficient for complex three-dimensional houses (roofs, dormers, valleys, soffits, projections, occlusions).

Decision: 10-step calibrated registration pipeline. Estimate thermal-camera extrinsics against RGB + property coordinate system → reproject radiometric data onto validated 3D surfaces → depth and occlusion testing → reject reflected, grazing, poorly registered surfaces → store per-region confidence. Local homography permitted only for verified planar regions.

Consequences: Thermal findings become scientifically defensible. Registration residuals are stored per region. Findings above configurable error thresholds require human review before Passport append. Colored thermal texture is never presented as a scientifically validated thermal finding.

ADR-003 · Preserve Original Sensor Data

Status: Accepted (v1.1)

Context: Normalized derivatives (TIFF, PNG, glTF) cannot be the source of truth. Conversion loss and lossy re-encoding invalidate any later scientific claim.

Decision: The **canonical mission package** stores original DJI radiometric files, RGB, video, flight logs, telemetry, sensor metadata, checksums, calibration data, environmental records, and operator notes. All derived assets are labeled as derivatives.

Consequences: Retention costs increase (mitigated by §13 lifecycle policy). Evidence chain-of-custody becomes provable. Reprocessing with a swapped `ThermalAnalysisProvider` or `PhotogrammetryProvider` is possible against the original evidence.

ADR-004 · Air Intelligence Scope Correction

Status: Accepted (v1.1)

Context: Exterior thermal imagery cannot establish indoor air quality, mold presence, CO_2, radon, or definitive airflow rates.

Decision: AWE Air scope initially limited to ventilation indicators, exterior air-leakage indicators, condensation-risk indicators, crawlspace/foundation ventilation observations, and areas requiring interior verification. Any indoor air quality diagnosis requires appropriate sensors and field validation.

Consequences: Product marketing must not claim indoor air quality. "A mold-risk indicator is not a mold diagnosis" enshrined in v1.1. Future indoor air quality capability requires separate sensor product + hardware.

ADR-005 · Risk-Tiered Human QA

Status: Accepted (v1.1)

Context: Blueprint v1.0 required human approval on nearly every finding — an operational bottleneck at scale.

Decision: Four-tier QA framework. Tier 1 auto-publishes if confidence and inputs pass. Tier 2 requires contractor/inspector. Tier 3 requires MFA-authenticated qualified human. Tier 4 requires licensed professional. Finding schema mandates `risk_tier`, `required_reviewer_role`, `release_restriction`.

Consequences: Operating cost scales sensibly. High-consequence findings still receive expert sign-off. Every publisher (Passport append, report render, Habitat sync) checks tier before releasing.

ADR-006 · Public Passport Access Is Not Hash-Based

Status: Accepted (v1.1)

Context: A permanent hash is not a bearer credential. Anonymous read access via hash-only leaks all passport data to anyone who ever received the URL.

Decision: Public access requires expiring signed links + scoped audience projection + revocation + access logging + rate limiting + optional recipient binding. Separate projections per audience (public / homeowner / contractor / adjuster / insurer).

Consequences: Existing `/passport/:hash` route must be re-routed through a signed-link factory during Phase 1a-2 migration. Homeowner-share, contractor-share, and adjuster-share flows all mint scoped links rather than exposing the hash.

ADR-007 · Optimistic Concurrency with Explicit Rebase

Status: Accepted (v1.1)

Context: "First-writer-wins" language implies silent data loss for the losing submission.

Decision: Optimistic concurrency with 7-step rebase workflow. Losing submissions preserved in `passport_sync_log`, automatically rebased when non-conflicting, routed to Human QA queue when materially conflicting. Never silently discarded.

Consequences: Cosign receipt scheme unchanged. `409 SEQUENCE_CONFLICT` still returned to client, but with a rebase-attempt link + queue routing on materially conflicting content.

ADR-008 · Layered Digital Twin Package

Status: Accepted (v1.1)

Context: A single glTF cannot be both a scientifically valid measurement record and an optimized mobile presentation.

Decision: Four canonical layers: Evidence (originals + reconstruction report), Measurement (validated surfaces + tolerances), Presentation (compressed glTF/KTX2 for mobile), Interchange (IFC/DXF/PLY/LAS/LAZ/E57/GeoTIFF only where source supports).

Consequences: Presentation optimization can never alter the measurement record. IFC/DXF exports are provably restricted to observed geometry — no invented concealed framing.

ADR-009 · Durable Delivery for Critical Events

Status: Accepted (v1.1)

Context: FastAPI background tasks lose work on process restart, deploy, or worker failure. Passport and Habitat delivery are business-critical.

Decision: Transactional outbox + durable jobs collection with locking and idempotent retries for the critical event set. In-process background tasks retained only for non-critical (UI notifications, cache warming). No external broker required in Phase 1.

Consequences: Approved findings pending Passport append, passport-to-Habitat handoffs, and evidence-package finalization survive deployment. External broker (NATS/Redis Streams) may be introduced in Phase 2+ if cross-service messaging demands it — not before.

ADR-010 · Validation Program Precedes Marketing Language

Status: Accepted (v1.1)

Context: Marketing phrases like "engineering-grade" or "insurance-grade accuracy" without empirical validation are legally and ethically indefensible.

Decision: Formal field-validation program (15 dimensions) defined before any external claim of accuracy. Truth Score displayed only in 10-point bands until calibrated. Product copy uses "Prototype scoring — not calibrated to industry standard yet" during pre-validation.

Consequences: Marketing team receives a documented gate. Any external claim of accuracy is tied to a specific validation report by version. Insurance carriers can request the validation report before granting evidentiary weight to Stratex outputs.

Meta-ADR · Documentation-Only Discipline

Status: Accepted throughout v1.1 revision pass

Decision: No implementation code was written during the v1.1 revision. No production deploy occurred. All v1.0 artifacts preserved read-only at `/app/memory/_v1.0_PRESERVED_*`.

Consequences: Reversibility is absolute. If v1.1 is rejected, v1.0 is untouched and can resume as the reference document.

ORIGINAL v1.0 REVIEW APPENDIX (Q1-Q14)

Preserved verbatim for reference. Answers superseded where v1.1 §§10–17 address them.

STRATEX CORE — NEXTGEN ARCHITECTURE REVIEW APPENDIX

Companion document to NEXTGEN_ARCHITECTURE_BLUEPRINT.md v1.0

Status: Answering Executive Review Questions 1-14

Author: TC · Stratex AI Product Manager, under directive of Anthony Cross

Purpose: Detailed responses to executive architecture review questions issued during the Blueprint hold.

Q1 — TWENTY-FOUR-AGENT ARCHITECTURE (justification)

Correction of framing

The blueprint's "24 agents" is a **logical software role count**, not a deployable-service count. In production, these logical agents collapse into **5 deployable services** during early phases:

SERVICE	LOGICAL AGENTS INSIDE	DEPLOYMENT
Mission Ops Service	Mission Commander · Mission QA · Flight Operations · Mission Assurance · Capture Validation	1 container, sync + background
Vision Intelligence Service	Photogrammetry · CAD/BIM · Digital Twin · Roof · Window · Door · Damage Detection · Measurement · Thermal Analysis	1 container, GPU-optional, async job queue
AWE Intelligence Service	Water · Air · Energy	1 container, async job queue
Estimating Service	Materials · Labor · Estimating · Maintenance Priority	1 container, sync
Delivery & Sync Service	Reporting · Passport Sync · Habitat Sync	1 container, sync + retry queue

Rationale: separately deploying 24 microservices during the prototype phase would introduce

network complexity, deployment cost, and observability burden with no operational payoff.

Per-agent detail table

#	AGENT	RESPONSIBILITY	INPUTS	OUTPUTS	MODEL / SERVICE	LOGICAL VS DEPLOYED	SYNC/ASYNC	HUMAN GATE	COST TIER
1	Mission Commander	Compose + validate mission plan	property_id, target_systems	mission_plan.json	Claude Sonnet + rule engine	Logical inside Mission Ops	Sync	Yes (dispatcher)	Low
2	Mission QA	Enforce 20-item preflight gate	mission_plan, drone_state, weather	gate_result{pass,fail,blockers[]}	Deterministic checks	Logical	Sync	No (all-or-nothing)	Free
3	Flight Operations	Live telemetry watch; safe-return decisions	telemetry stream	control_decisions[]	Rule engine	Logical	Async streaming	Escalates	Low
4	Mission Assurance	In-flight GSD/overlap QA	live imagery metadata	additional_shots[]	Statistical + Claude	Logical	Async streaming	No (auto-retake)	Low
5	Capture Validation	Radiometric integrity, RGB↔thermal registration	asset bundle	validation_report	Statistical + Claude	Logical	Async job	Yes	Low
6	Photogrammetry	Point cloud + mesh generation	RGB assets	.glb, .obj, point_cloud	**AWAITING INTEGRATION** (Pix4D / RealityCapture / OpenDroneMap)	Deployed later	Async job	Yes	High
7	Thermal Analysis	Radiometric anomaly detection	thermal R-JPG assets	thermal_findings[]	Statistical + Claude	Logical	Async job	Yes	Medium
8	CAD/BIM Generation	Structural layers, framing extraction	point cloud + mesh	.ifc, .rvt derivatives	**AWAITING INTEGRATION** (commercial)	Deployed later	Async job	Yes	High
9	Digital Twin	Fuse 3 layers into single twin	reality mesh + CAD + intelligence	twin manifest	Deterministic composition	Logical	Async	Yes	Low
10	Roof Intelligence	Plane, facet, damage classification	RGB + twin	roof_findings[]	Claude + CV heuristics	Logical	Async	Yes	Medium
11	Window Intelligence	Type, dimensions, glass, leakage	RGB + thermal + twin	window_findings[]	Claude + CV	Logical	Async	Yes	Medium
12	Door Intelligence	Type, material, seal condition	RGB + thermal + twin	door_findings[]	Claude + CV	Logical	Async	Yes	Medium
13	Water (AWE)	Moisture retention, water risk	thermal + roof geometry + weather history	water_findings[], water_score	Claude + rules	Logical inside AWE Service	Async	Yes	Medium
14	Air (AWE)	Ventilation performance	thermal + door/window findings	air_findings[], air_score	Claude + rules	Logical	Async	Yes	Medium
15	Energy (AWE)	Envelope performance, heat-loss score	thermal + envelope + weather	energy_findings[], energy_score	Claude + rules	Logical	Async	Yes	Medium
16	Damage Detection	Cross-system anomaly clustering	all findings[]	prioritized_damage[]	Claude	Logical	Async	Yes	Medium
17	Measurement Intelligence	Extract precise measurements from twin	twin + registration	measurements[] with tolerance	Deterministic geometry	Logical	Sync	Yes	Free
18	Materials Intelligence	Compose BOM	measurements + roof/window/door findings	bom[]	Catalog lookup + Claude	Logical inside Estimating	Sync	Yes	Low

#	AGENT	RESPONSIBILITY	INPUTS	OUTPUTS	MODEL / SERVICE	LOGICAL VS DEPLOYED	SYNC/ASYNC	HUMAN GATE	COST TIER
19	Labor Intelligence	Compose labor plan	bom + regional rates	labor_plan	Catalog lookup + Claude	Logical	Sync	Yes	Low
20	Estimating Intelligence	Final estimate w/ waste, tax, O&P	bom + labor + preferences	estimate.json	Deterministic math	Logical	Sync	Yes	Free
21	Maintenance Priority	Rank issues by urgency + cost avoidance	all findings + estimate	priorities[]	Composite scoring	Logical	Sync	Yes	Free
22	Reporting Intelligence	Compose enterprise report	approved intelligence bundle	report.pdf + report.json	Template + Claude narrator	Logical inside Delivery	Sync	No (post-approval)	Low
23	Passport Sync	Emit ledger delta to Passport	approved bundle	sync receipt	Deterministic	Logical	Async + retry queue	No	Free
24	Habitat Sync	Emit homeowner packet to Habitat	approved bundle	sync receipt	Deterministic	Logical	Async + retry queue	No	Free

Combination rationale

Agents 13-15 (AWE) cannot be combined because Water, Air, and Energy each require distinct

input signals (weather history vs door/window vs envelope) and produce independently-reported scores.

Similarly, Roof/Window/Door intelligences cannot be combined because approval authority

differs (a contractor may approve roof findings but require an engineer for structural doors).

Q2 — PROPERTY PASSPORT AUTHORITY

Ownership model (definitive)

CONCERN	AUTHORITY
Creates Passport	Stratex Passport service — issued via `POST /passport/v1/mint` with a nonce from Core
Owns canonical identifier	Stratex Passport service — 12-char uppercase hash, globally unique
May append records	**Stratex Passport service only.** Core submits deltas via `/passport/v1/append`; Passport service re-verifies chain integrity before writing
May correct records	Nobody. Records are immutable. Corrections are new appends with an explicit `supersedes` pointer to the prior record
Versioned via	Sequential ledger `seq` + SHA-256 chain (`prev_hash` field) — already implemented today for the native passport
Findings superseded via	New ledger entry of type `SUPERSEDE` carrying `prior_seq`, `reason`, and `superseding_finding_id`. Historical record is never destroyed
Core submission path	`POST /passport/v1/append {passport_id, delta_bundle, core_signature}` → Passport verifies + writes + returns cryptographic receipt
Habitat receipt path	`GET /passport/v1/projections/homeowner/:passport_id` — a read-only projection scoped to homeowner-appropriate fields
Conflict resolution	**First-writer-wins with per-property sequence numbers.** Concurrent appends fail with `409 SEQUENCE_CONFLICT`, forcing the loser to re-read and re-submit
Cryptographic proof	Every accepted append returns a SHA-256 receipt computed over `{payload, seq, prev_hash, signed_at}`. Core stores this receipt in its `sync_audit_log`
Deployment	**Standalone service** post-parity. Today it is embedded in `/app/backend/routes/passport.py` — this is the first module extracted in Phase 2

Access-control matrix

ACTOR	READ	APPEND	CORRECT	HABITAT RECEIVE
Contractor (Core)	✅ (their org's properties)	✅ (approved bundles only)	❌	❌
Homeowner	✅ (their property only, homeowner projection)	❌	❌	✅
Adjuster	✅ (via magic link)	✅ (cosign entries only — already exists)	❌	❌
Passport service itself	✅	✅ (all writes go through it)	✅ (via SUPERSEDE semantics)	—
Habitat service	❌ direct	❌	❌	✅ pull only

Existing Claim Snapshot cosign receipt pattern (§8 blueprint) is the canonical template for how every future append works.

Q3 — PROVENANCE INTERFACE

Visual-noise avoidance rules

Always visible (chip in-line) — applies to high-consequence numbers only:

- Envelope Score
- Moisture %
- Repair Estimate $
- AWE™ scores (Air, Water, Energy)
- Truth Score
- Confidence %

Progressive disclosure — for supporting values:

- Every measurement (dimension, count, area) shows a small colored dot only
- Tap / hover reveals a hover-card with the full provenance object
- "Open Evidence" button in the hover-card drops into an Evidence Drawer

Mobile behavior:

- Chips shrink to a single colored dot (4px) on ≤430 px viewports
- Long-press opens the provenance drawer (not hover)
- Drawer takes 90% of screen height, dismissible with swipe-down

Evidence Drawer contents:

- Source asset thumbnail (with a "view full size" affordance)
- Model + version + prompt version
- Confidence, timestamp, signer (if human-approved)
- "Reject / Escalate" affordance if the user has QA role

Provenance vs Confidence vs Truth Score (clarified)

CONCEPT	QUESTION IT ANSWERS
Provenance	*Where did this value come from?* — Source lineage
Confidence	*How sure is the model / measurement?* — Per-agent numeric or human-verified
Finding confidence	Aggregate confidence across all inputs for a single finding
Truth Score	*How trustworthy is this entire property record?* — See Q4

Q4 — TRUTH SCORE (honest specification)

Definition (revised, non-marketing)

Truth Score is a **composite property-record trust indicator**, expressed on a 0-100 scale in **10-point bands** (never in decimal form) with descriptive language:

BAND	LANGUAGE
90-100	**Field-verified · production-grade**
70-89	**AI-assisted · human-approved**
50-69	**AI-assisted · pending human review**
30-49	**Partial coverage**
0-29	**Insufficient evidence**

Component inputs

1. **Evidence coverage** (0-30 pts) — fraction of expected asset types present (RGB, thermal, dimensional)
2. **Model confidence average** (0-25 pts) — mean confidence of all findings
3. **Human verification ratio** (0-25 pts) — % of findings signed by human QA
4. **Recency** (0-10 pts) — age of freshest scan
5. **Cross-registration** (0-10 pts) — RGB↔thermal alignment score (if AWE performed)

Weighting

Deterministic weighted sum with published coefficients — never a model output.

Missing-evidence behavior

Missing evidence subtracts from Evidence Coverage. Never treated as "confident zero."

Contradictory-evidence behavior

Contradictions between two findings trigger a **CONTRADICTION** event that:

- caps Truth Score at 60 until reconciled
- surfaces both findings to the Human QA queue

Human verification effect

Each human-approved finding contributes to the Human Verification Ratio component. A property with 100% human approval on all findings can reach the 90-100 band.

Versioning

Truth Score has its own version number (e.g. `TS.v1.0`). Historical scores are frozen in the passport ledger.

Calibration

Requires empirical field-verification runs against ground truth before we publish the score externally. Until then, the score is displayed with the label **"Prototype scoring — not calibrated to industry standard yet."**

Display language rules

- **Prohibited**: "97.43%," "99.7% accurate," "AI-verified truth."
- **Required**: 10-point band + descriptive label + provenance chip

Marketing restrictions

- Truth Score may **not** be presented to insurance carriers or in contracts as a warranty of accuracy until calibration is published.
- May be presented **only** as a "record-completeness signal" during the prototype phase.

The four levels distinguished

LEVEL	SCOPE
Model confidence	Per-inference (one AI call)
Measurement confidence	Per-measurement (one dimension)
Finding confidence	Per-finding (composite of contributing model + measurement confidences)
Truth Score	Per-property record (composite of all findings + coverage + verification)

Q5 — AWE™ MISSION SEPARATION

Recognized: AWE requires a separate mission

The blueprint is amended: AWE™ is **not a processing toggle on daytime imagery**. Three explicit mission products exist:

Product 1 · Stratex Core DayScan™

- **When**: daytime, dry conditions
- **Captures**: RGB imagery only
- **Produces**: measurements, roof/exterior geometry, damage imagery, Digital Twin (Reality + CAD layers), material quantities, DayScan report
- **Updates**: Property Passport DayScan chain

Product 2 · Stratex AWE™ Scan

- **When**: nighttime, only when environmental eligibility passes:
- ΔT between indoor and outdoor ≥ 15°F (configurable per region)
- No rain within last 4 hours
- Wind < 15 mph
- Humidity < 80%
- No direct solar loading (≥1 hour past sunset)
- **Captures**: radiometric thermal + reference RGB
- **Produces**: Air / Water / Energy findings with false-positive controls (moisture-mimic exclusion, ventilation-shadow rejection)
- **Requires**: field verification for any finding above a materiality threshold before Passport entry
- **Updates**: Property Passport AWE chain

Product 3 · Stratex Elite™ Property Intelligence

- Combines DayScan + AWE Scan (with cross-registration between RGB and thermal datasets)
- Produces unified evidence graph, combined report, **AWE Index™** (composite Air/Water/Energy)
- Updates Property Passport with a single approved bundle referencing both scan hashes

Mission planner enforcement

The Mission Planner (§3 blueprint) must present the operator with a **product selector** as the first step: DayScan vs AWE Scan vs Elite. The 20-item ATC gate then loads the product-specific validation subset.

Q6 — DIGITAL TWIN ARCHITECTURE

Canonical decisions

CONCERN	DECISION
Coordinate system	WGS84 lat/lon + local ENU (East-North-Up) origin per property; twin is stored in local ENU with a lat/lon origin pin
Geometry source	Photogrammetry point cloud + mesh (from RGB) — **AWAITING INTEGRATION**
RGB texture source	DayScan RGB assets
Thermal-to-RGB registration	Feature-based homography per capture; per-pixel registration confidence stored
Layer manifest	JSON manifest listing each layer (Reality, CAD, Intelligence) with URIs, hashes, version
Formats	Storage: glTF 2.0 (`.glb`) canonical; IFC 4 and DXF for CAD; PLY for point clouds; radiometric TIFF for thermal source
Measurement provenance	Every extracted measurement carries `derived_from: [asset_ids]` + tolerance ± value
Level-of-detail	glTF LOD hierarchy: L0 (full), L1 (medium), L2 (mobile); browser streams L2 by default, promotes to L0 on zoom
Version comparison	Two twin versions can be diffed; findings that shift receive a `DIFF` event
Browser streaming	glTF over HTTPS with Draco compression + KTX2 textures
Mobile performance	L2 mesh + 1k textures target < 20 MB total; renders on iPhone 12 and later at 60 fps
Vendor-neutral storage	S3-compatible object storage. No vendor-specific format inside the canonical bundle
Exportability	Every twin can be exported as `.glb` + `.ifc` + `.dxf` bundle to a customer download link
External processor replacement	Photogrammetry, CAD/BIM, and radiometric processing are all **adapter contracts**. Concrete provider (Pix4D, RealityCapture, OpenDroneMap) is swappable without changing Core

Stratex-owned vs external-processor-owned

Stratex owns forever:

- Mission package (plan, waypoints, targets)
- Evidence graph (which asset produced which finding)
- Findings + approvals + Passport records
- The layer manifest and its version history
- Signed measurement provenance

External processors own only the raw mesh/CAD generation step and are treated as replaceable adapters behind a `PhotogrammetryProvider` / `BimProvider` interface.

Q7 — DATA-STATE HONESTY

Adopted implementation-state vocabulary

Every feature, integration, and displayed value shall carry exactly one of:

STATE	MEANING
Operational	Live end-to-end in production, verified by tests
Partially operational	Some flows live, others use fallback or fixtures — clearly labeled which
Mocked	Fixture data; nothing real behind it
Planned	Design complete, not built
Awaiting credentials	Built but requires API key / license not yet obtained
Awaiting hardware	Built but requires drone / sensor / dock not yet on hand
Awaiting validation	Built but requires field calibration before trust
Deprecated	Phasing out; do not use for new work
Legacy	Frozen, read-only, under `/legacy/*`

Marketing rules

- No screen may claim "operational" for a component that is any other state
- The UI shall render each state with a distinct chip color and label
- Reports shall footer with a state summary of every value in the document

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Q8 — EXISTING ASSET PRESERVATION MAP

ASSET	DECISION	RATIONALE
Official Stratex branding (logo, colors, typography)	**Preserve unchanged**	Approved; carried into NextGen shell
Property Passport ledger (hash-chained)	**Preserve unchanged**, migrate storage to `passport_ledger` collection with Pydantic model	Working crypto — do not touch the algorithm
Claim Snapshot engine	**Refactor** into `findings` + specialized `claim_snapshot` view; keep public API	Genuine value; needs normalized storage
Adjuster co-sign SHA-256 receipt	**Preserve** as the canonical template for all future human sign-offs	Generalize the pattern to every Human QA event
Open-Meteo weather integration	**Preserve unchanged**; extract into `WeatherProvider` adapter	Simple, working, free
Live Ops WebSocket bus	**Refactor** onto formal event bus (§7 blueprint); keep client contract	WS is fine; server logic needs event log
PDF / Report pipeline (Playwright + cache-first)	**Preserve**; graduate to Report Studio in Delivery Service	Production-safe already
Reports Binder (PyMuPDF)	**Preserve** as embedded viewer in Report Studio	Works
Existing authentication (JWT + bcrypt, CEO login)	**Refactor** — extract to Identity Service, expand to full RBAC	Foundation good; scope must grow
Existing authorization	**Refactor** — introduce `require_auth` + `require_org` on all product routes	Critical gap today
Service credentials / HMAC patterns	**Preserve** patterns; regenerate secrets on each service split	Reusable
Existing Core → Passport → Habitat contracts	**Migrate** — Passport becomes standalone; Habitat contract is new	Formalized in Q2
Existing test suites (`test_iter17_new_features.py` … `test_iter21_cosign.py`)	**Preserve** as legacy regression harness; add NextGen suite alongside	Keep the safety net
Static illustrations in `/twin/*` and `_binder_cache/*`	**Retire** once real twin renders exist, but keep for the interim label as `DEMO DATA`	Cannot be shown as real intelligence
`_sample_analysis()` hardcoded fixture	**Retire** — replaced by the AI Workforce pipeline	Explicit deletion at Phase 3

Nothing is deleted through omission. Every asset is accounted for.

Q9 — SECURITY SCOPE (phased introduction)

Phase 1 (Foundation)

- **Initial roles**: `admin`, `contractor`, `pilot`, `inspector`, `homeowner`, `adjuster`
- **Initial permissions**: read/write/approve/sign
- **Organization isolation**: `org_id` required on every mutable resource; enforced in query builder
- **Property-level access**: `property_org_bindings` table; contractors see only their org's properties
- **Contractor access**: read all properties in their org, write findings after inspector approval
- **Homeowner access**: read-only projection of their property only
- **Pilot / operator access**: read mission plans, write telemetry + capture assets
- **Administrator access**: full within their org; cross-org requires service token
- **Engineer/adjuster reviewer access**: read + sign findings/reports via magic-link tokens (existing cosign pattern)

Service-to-service authorization

- mTLS between Core ↔ Passport ↔ Habitat (Phase 2+)
- Short-lived service tokens with per-endpoint scopes

Audit logging

- Every mutation → `audit_events` (actor, action, resource, before, after)
- Retention: 2 years hot, 7 years cold storage — **AWAITING INTEGRATION**

Secrets management

- All secrets in `.env` today
- Move to Emergent-managed secret store or HashiCorp Vault (Phase 5)

Rate limiting

- Per-user + per-IP token buckets on all public endpoints
- Passport, cosign, storm APIs: 60 req/min per IP; 300 per authenticated user

File access

- All object-storage assets served via **signed URLs** with expiry (default 5 min)
- Signed uploads via presigned POST
- Every download logged to `asset_provenance`

Explicit Phase 1 stops

- No SSO in Phase 1
- No field-level encryption in Phase 1 (Phase 3)
- No SIEM in Phase 1 (Phase 5)

Q10 — DATABASE-DOMAIN JUSTIFICATION

Correction: the 10 domains described in §4 of the blueprint are **logical modules**, not 10 separate databases. During prototype and MVP phases, all 10 domains live inside a **single MongoDB database** with strict per-collection schemas and boundaries enforced at the application layer.

Per-domain table

DOMAIN	PURPOSE	CORE ENTITIES	OWNERSHIP BOUNDARY	TRANSACTIONS	RETENTION	SENSITIVITY	SEPARATE DB?
Identity & Access	Auth + RBAC + audit	orgs, users, roles, api_keys, audit_events	Identity Service	ACID-lite (Mongo multi-doc tx)	7 yrs (audit)	HIGH (PII)	No — logical only
Property Registry	Canonical property record	properties, addresses, parcels, owners, contractors	Property Service	Multi-doc tx on write	Permanent	MEDIUM (PII)	No
Mission Operations	Mission lifecycle	missions, mission_plans, mission_validations, mission_telemetry, capture_sessions	Mission Ops Service	Sequential per mission	3 yrs	LOW-MEDIUM	No
Capture Assets	Binary asset metadata (binary in S3)	assets, asset_derivatives, asset_provenance	Vision Service	Metadata only	7 yrs (bodies in S3)	MEDIUM	No (metadata) / Yes (S3 binaries)
Intelligence Findings	AI + human findings	findings, finding_evidence, finding_confidence, agent_runs	Vision + AWE + Estimating Services	Append-only ledger-like	Permanent	MEDIUM	No
AWE™ Metrics	AWE readings + scores	awe_readings, awe_scores, awe_history	AWE Service	Sequential per property	Permanent	LOW	No
Estimating	Materials + labor + estimates	materials_catalog, labor_catalog, estimates, estimate_versions	Estimating Service	Versioned; last-write-wins	Permanent	LOW-MEDIUM (business)	No
Reports & Deliverables	Rendered reports	report_templates, reports, report_versions, report_signatures	Delivery Service	Sequential	Permanent	LOW	No
Passport & Habitat Sync	Sync audit + queue	passport_ledger, passport_sync_log, habitat_sync_log, sync_retry_queue	Passport Service	Ledger append + idempotent queue	Permanent	HIGH (cryptographic)	**Yes** — Passport becomes standalone in Phase 2
Compliance & Observability	Provenance, events, cost	provenance_chain, event_log, error_events, cost_ledger	Platform	Append-only	2 yrs hot	LOW	No

Only the Passport domain justifies extraction into a separate database (Phase 2) — because its records are the immutable source of truth for all downstream systems. Everything else remains logical modules inside the primary Mongo database until scale demands otherwise.

Q11 — API-PLANE JUSTIFICATION

Correction: the 5 planes are **logical route-prefix conventions with distinct middleware chains**, not 5 separate network services. All planes run inside the same FastAPI app initially; some may split later.

Per-plane table

PLANE	PREFIX	CONSUMERS	AUTH	AUTHZ	VERSIONING	RATE LIMITS	ERRORS	IDEMPOTENCY	AUDIT	NETWORK BOUNDARY?
Public Read	`/api/pub/v1/`	Anonymous (via passport hash)	Signed URL / hash	Resource-level	URL versioned	60/min/IP	RFC 7807 JSON	GET only	Read log to `audit_events`	No (same process, edge-cacheable)
Product	`/api/v1/`	Authenticated users	JWT	RBAC + ABAC (org, property, role)	URL versioned	300/min/user	RFC 7807	Idempotency-Key header on POST/PUT	Full audit	No (same process)
Agent	`/api/agents/v1/`	Backend agent runners	Service token	Scope-per-endpoint	URL versioned	Internal only	RFC 7807	Idempotency-Key required	agent_runs table	**Yes** — internal network only, blocked at ingress
Sync	`/api/sync/v1/`	Passport, Habitat services	mTLS + service token	Service allow-list	URL versioned	Bounded by queue depth	RFC 7807 + retry semantics	Idempotency-Key required	Full audit	**Yes** (Phase 2+)
Webhook	`/api/hooks/v1/`	External inbound (DJI, FAA, weather)	HMAC signature	Provider allow-list	URL versioned	Provider quotas	202 on accept, HMAC failure → 401	Idempotency by event id	Full audit	No (same process)

Actual network boundaries introduced only where necessary — Agent and Sync are gated at the ingress layer. Everything else is logical.

Q12 — EVENT-DRIVEN ARCHITECTURE (Phase 1 events only)

Phase 1 minimum viable event set

Only these events are wired in Phase 1. All others are deferred until real workflows demand them.

EVENT	PRODUCER	CONSUMER	PAYLOAD	IDEMPOTENCY KEY	RETRY	FAILURE	DLQ	AUDIT	SYNC ALTERNATIVE?
`mission.planned`	Mission Ops	Mission Validation	mission_id, plan	mission_id	3× exp backoff	Log + surface in Mission Ops UI	Yes	audit_events + event_log	Sync is fine; keep async
`mission.validated`	Mission Validation	Mission Ops (unlock launch)	mission_id, pass/fail, gate_details	mission_id	none	Block launch	No	audit_events	Sync (immediate)
`capture.received`	Vision (adapter)	Vision Service	mission_id, asset_bundle_id	asset_bundle_id	3×	Escalate to Human QA	Yes	audit_events + event_log	Async needed (large uploads)
`finding.produced`	Any AI agent	Human QA	finding_id, agent_id, confidence	finding_id	none	Halt Passport append	No	agent_runs	Sync (blocks Passport)
`finding.approved`	Human QA	Passport Sync	finding_id, signer, receipt	finding_id	none	Halt sync	No	audit_events	Sync (blocks sync)
`passport.appended`	Passport Service	Habitat Sync	passport_id, seq, receipt_hash	(passport_id, seq)	3×	sync_retry_queue	Yes	passport_sync_log	Async (external system)
`habitat.received`	Habitat (webhook back)	Passport Sync	passport_id, seq, ack	(passport_id, seq)	—	Alert ops	Yes	habitat_sync_log	Async (ack)

No full event-bus infrastructure in Phase 1. These events are recorded in `event_log` and dispatched via in-process async tasks (FastAPI background tasks). If cross-service messaging becomes necessary in Phase 2+, we upgrade to a proper broker (NATS or Redis Streams) — not before.

Q13 — TECHNOLOGY DECISIONS (justified)

TECH	PROBLEM SOLVED	WHY EXISTING IS INSUFFICIENT	ADOPTION COST	MIGRATION RISK	SIMPLER ALTERNATIVE	NECESSARY IN PHASE
React Query	Server-state caching, retries, background refresh, optimistic UI	Current app uses ad-hoc `useEffect` + `fetch`; race conditions and stale data are already visible	Low (drop-in; ~1 day)	Low (per-page adoption)	Continue with useEffect (rejected — bugs)	Phase 1b
Zustand	Small global UI state (e.g. current mission, active twin layer)	React Context works but rerenders too much	Very low	None	Redux (too heavy); Context (already present)	Phase 1b
Event-driven backbone	Decoupling AI pipeline stages	Current code is deeply coupled (`storm_watcher` reaches into passport internals)	Medium	Medium	Sync in-process only (acceptable in Phase 1)	**Deferred to Phase 2**
Blue/green deployment	Zero-downtime cutovers	Emergent single-container deploys create brief unavailability	Low (Emergent may not support natively)	Low	Just rollback via checkpoint (fine for now)	**Deferred to Phase 5**
Feature flags	Ship stubs safely, control AWAITING INTEGRATION states	Currently commenting out code	Low (implement in-house)	None	Env vars (works but less granular)	Phase 1b (in-house key-value flag)
Cryptographic receipts	Prove what was submitted / approved	Already implemented (cosign) — extend the pattern	Free (already exists)	None	(already chosen)	Phase 1 (existing)
RBAC	Role-based access control	Current app has none on product routes	Medium (~1 week)	Medium (retrofit every route)	ACLs only (too rigid)	Phase 1 mandatory
ABAC overlay	Property/org-level access rules	RBAC alone can't gate by `property_org_binding`	Low on top of RBAC	Low	RBAC-only (rejected — leaks data)	Phase 2

Rejected in this round

- **Kubernetes multi-cluster**: unjustified until scale demands
- **Kafka**: overkill; NATS/Redis Streams sufficient when we need a broker
- **GraphQL**: REST + tight schemas is sufficient
- **Rust rewrite of any component**: no
- **Any new frontend framework**: React is the standard here

Q14 — PHASE 1 SCOPE CONTROL (explicit)

Phase 1 **creates the architecture and operating shell** but delivers **no finished production intelligence engines**.

Phase 1 SHALL establish (deliverable list)

1. Application boundaries (Production / NextGen / Legacy)
2. Navigation (14 modules, canonical routes)
3. Workflow state machine (14-stage mandatory chain)
4. Mission shell (Mission Ops workspace stubs)
5. Property shell (Property Portfolio + Property Detail with tabs)
6. Evidence schema (asset ↔ finding ↔ approval)
7. Provenance schema (chip component + backing data model)
8. Agent-run audit schema (`agent_runs` collection)
9. Passport contract (Q2 above)
10. Habitat synchronization contract (§8 blueprint + Q2)

11. Digital Twin layer contract (Q6)
12. AWE mission contract (Q5)
13. Design-system foundation (tokens, provenance chip, module frame)
14. Security foundation (RBAC skeleton, `require_auth` middleware)
15. Test strategy (regression harness plan)

Phase 1 SHALL NOT claim to deliver

- Production-ready photogrammetry
- Production-ready thermal diagnosis
- Automated CAD/BIM generation
- Field-validated AWE conclusions
- Automated damage classification with contract-grade accuracy
- Any measurement presented as production-truth without human sign-off

Every screen in Phase 1 delivering intelligence content shall wear one of these labels:

`DEMO DATA · MOCKED · AWAITING INTEGRATION · AWAITING HARDWARE · AWAITING VALIDATION`.

Success criteria (Phase 1 exit gate)

- All 14 items above delivered
- Blueprint + Appendix + Executive Review signed off
- Screenshots of every workspace captured on desktop + mobile
- Regression suite green
- Zero fabricated intelligence displayed in the shell

APPENDIX — REVISION HISTORY

VERSION	DATE	AUTHOR	CHANGE
1.0	Feb 26, 2026	TC	Initial draft of NextGen Architecture Blueprint (623 lines)
1.0-app	Feb 26, 2026	TC	Review Appendix v1.0 answering Executive Review Questions 1-14

End of Review Appendix.